

Anatomy and Physiology of Thalamus & Internal Capsule

Principles and Practices of Osteopathic Medicine I

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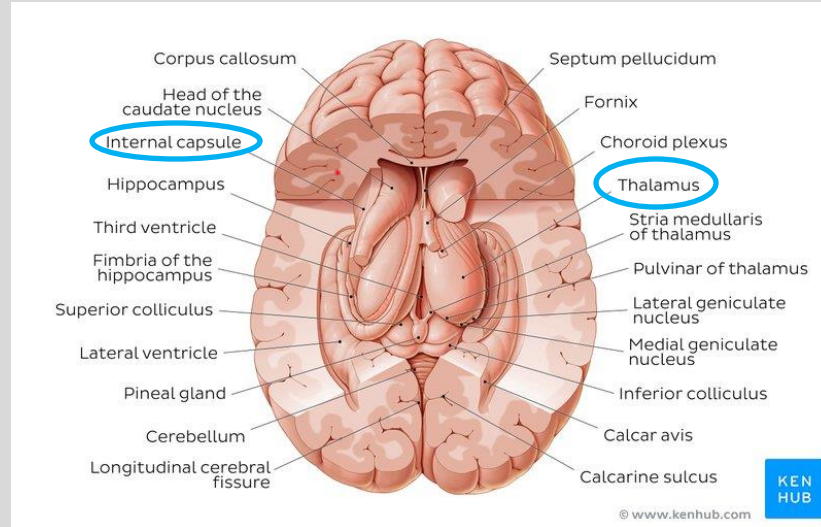
Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives

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1. Describe the gross **organization** and **connections** of the thalamus in relation to other structures in the brain.
2. Identify the **major nuclei** of thalamus associated with sensation, motor control, pain, and consciousness.
3. Differentiate the information carried by different segments of the **internal capsule**.
4. Understand the **blood supply** of thalamus and internal capsule.



See **detailed objectives** in CPG.
PLEASE READ THE Scholar Rx Brick!

Session Objectives Mapped to COMLEX Blueprint

- **Dimension 2**
 - **Clinical presentation 4.** Patient presentations related to the nervous system and mental health
 - 4.11 Sensory disturbances and pain
 - tactile disturbances, including sensory loss, numbness, vibration/temperature/proprioception loss, tingling, and paresthesia
 - pain, chronic nonmalignant
 - pain, neuropathic, nociceptive, mixed, sympathetic
 - nerve-, muscle-, and pain-related syndromes, including complex regional pain syndrome, post-herpetic neuralgia, meralgia paresthetica

Session Objectives Mapped to USMLE Blueprint

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V. Nervous System & Special Senses

A. Normal Processes

2. Organ structure and function

iv. Sensory systems

a. general sensory modalities, including sharp, dull, temperature, vibratory, and proprioception

B. Abnormal Processes: Health and Health Maintenance, Screening, Diagnosis, Management, Risks, Prognosis

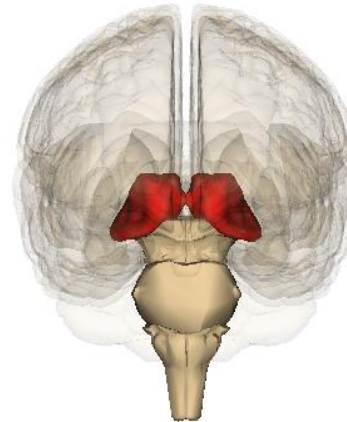
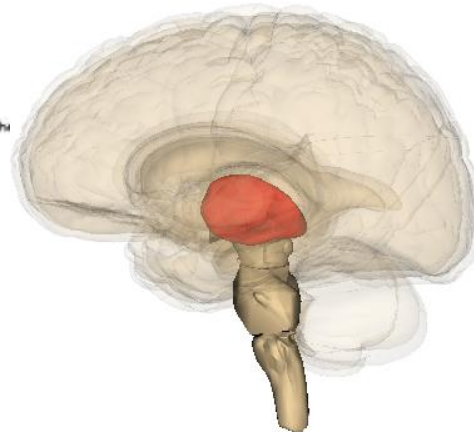
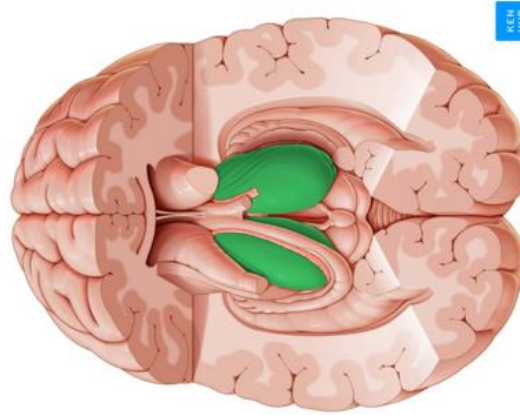
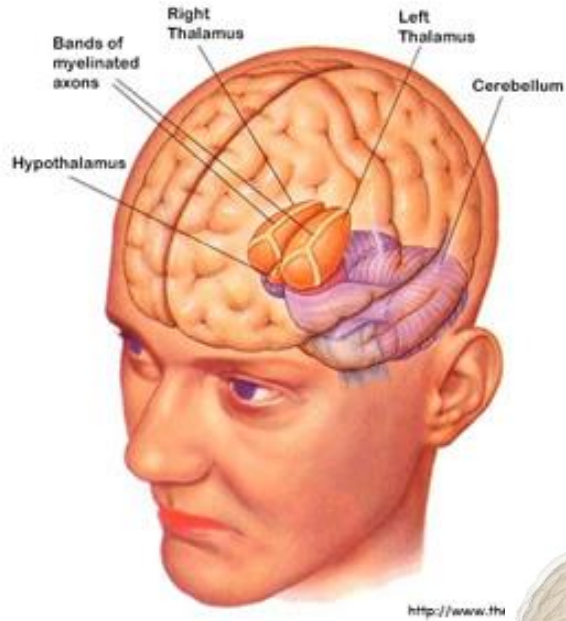
5. Cranial and peripheral nerve disorders

iii. Neurologic pain syndromes: complex regional pain syndrome (reflex sympathetic dystrophy, causalgia); fibromyalgia; postherpetic neuralgia; phantom limb pain/syndrome; thalamic pain syndrome; trigeminal neuralgia

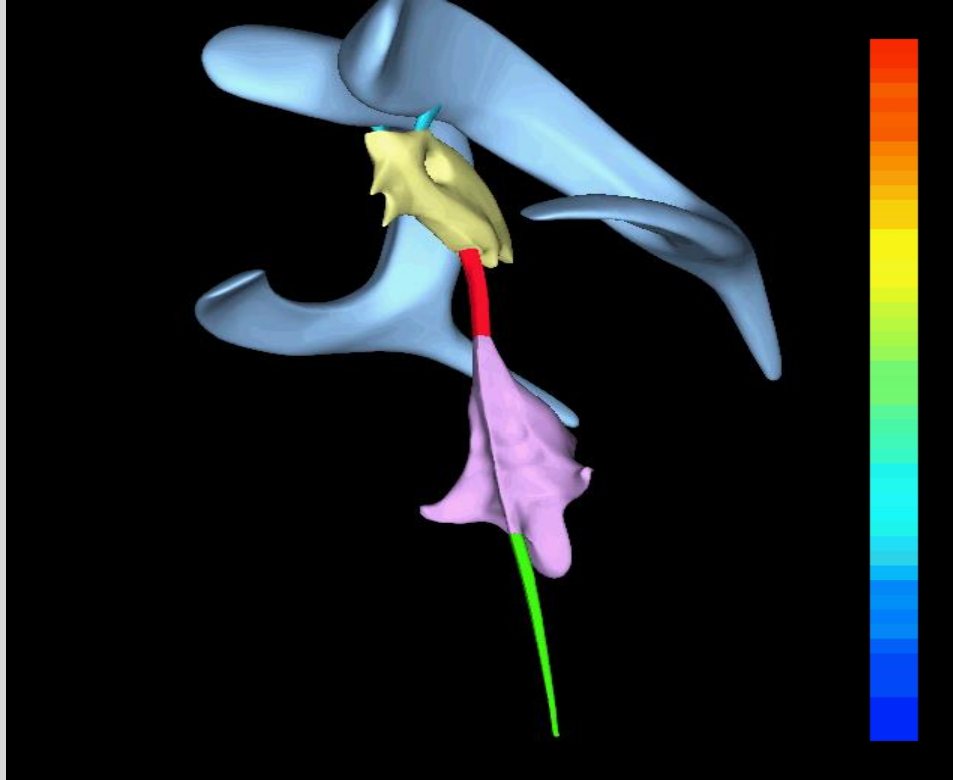
Thalamus

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3D Model of Ventricles



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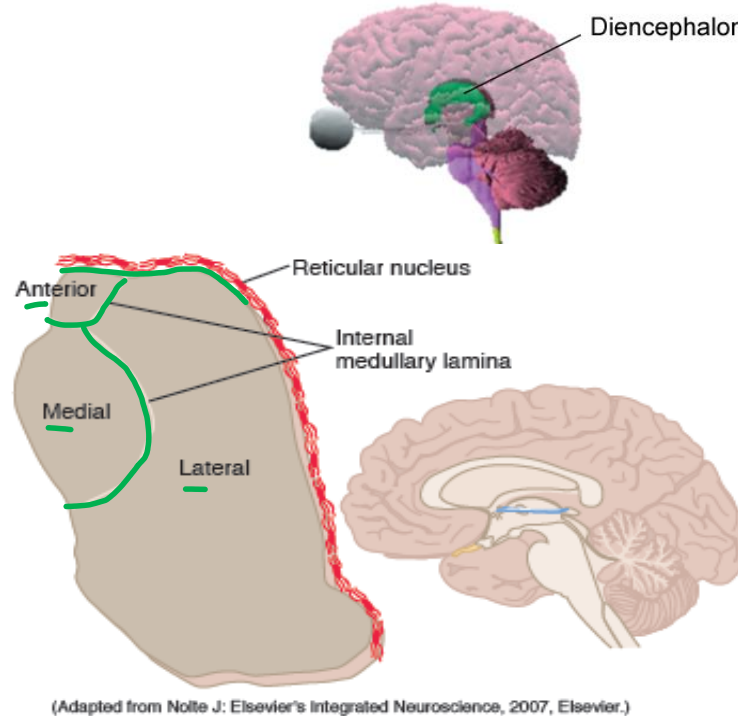
https://en.wikipedia.org/wiki/Ventricular_system#/media/File:Human_Ventricular_system_colored_and_animated.gif

Anatomy Overview

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- Thalamus is the **largest** component of **diencephalon**, lies between brainstem & cerebral hemisphere
- Almost all thalamic nuclei have rich **reciprocal connections** with **cerebral cortex**
- Thalamus is divided into 3 principal nuclear masses (**anterior, medial, lateral**) by the **internal medullary lamina** (also contains intralaminar nuclei)
- Thin **reticular nucleus** lies on lateral aspect of thalamus



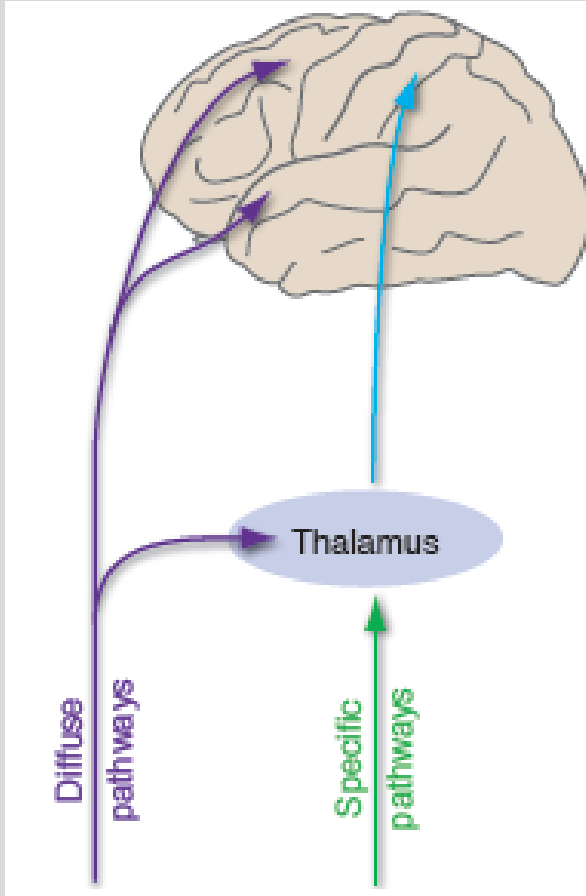
Functional Overview

- Nuclei that transmit general and special **sensory information** corresponding to regions of **sensory cortices**
- Nuclei that receive impulses from **cerebellum** and **basal ganglia** and interface with **motor regions of frontal lobe**
- Nuclei that have connections with **associate** and **limbic areas** of cerebral cortex

Inputs to Cortex

Cerebral cortex receives inputs via:

- 1) Thalamus sends **specific** information to restricted cortical areas via **relay nuclei**
- 2) **Diffuse, modulatory** inputs that set the general level of excitability of broad cortical areas via **association nuclei**



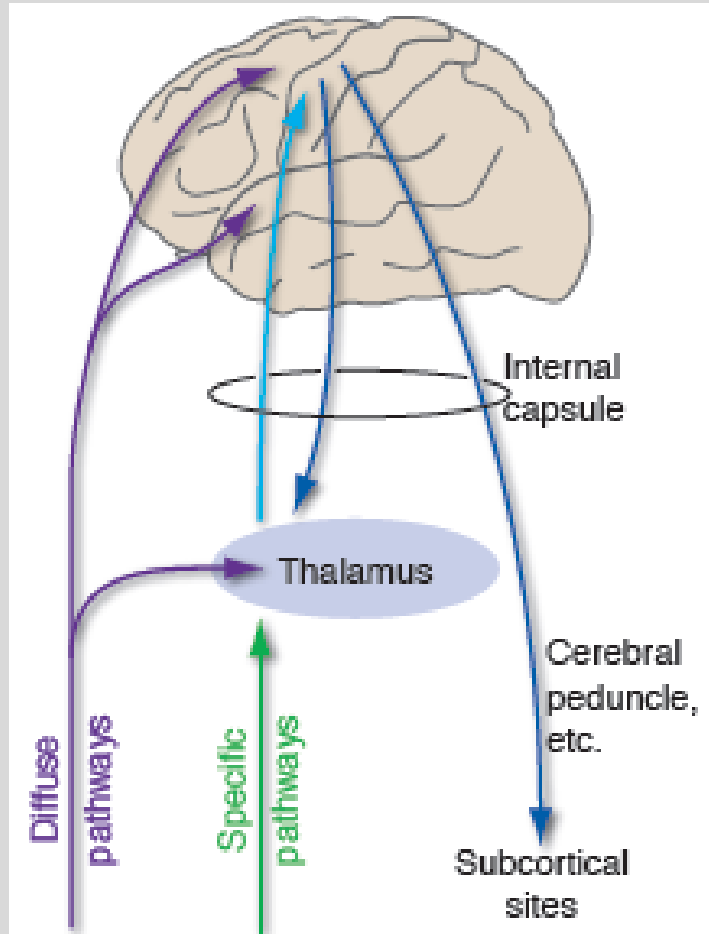
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Afferent vs. Efferent Pathways

A fundamental difference between pathways entering the cortex and pathways leaving the cortex:

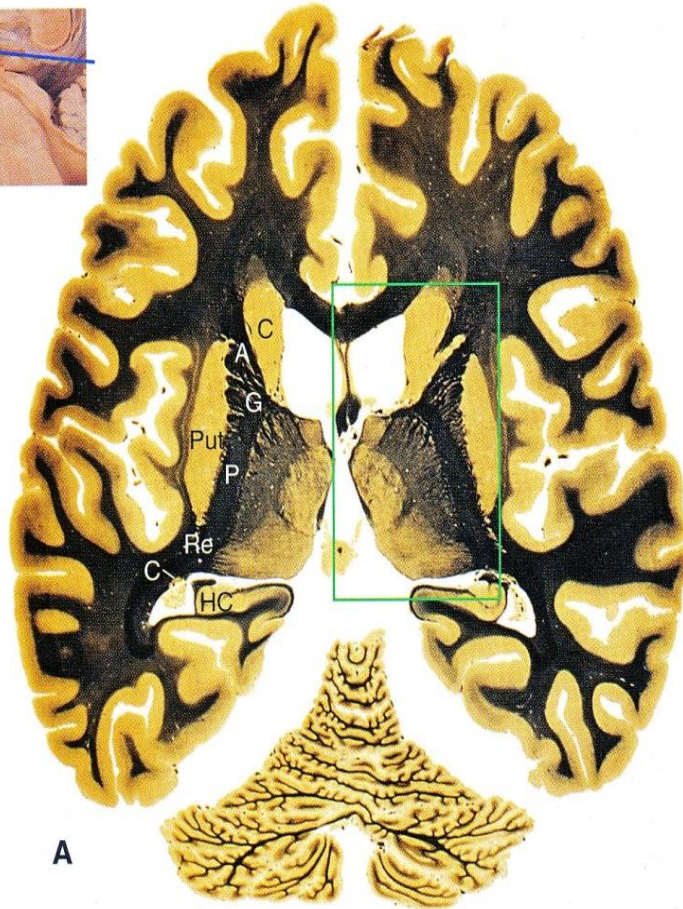
Efferent pathways (e.g., the corticospinal, corticobulbar and corticopontine tracts) can go directly from the cortex to their destinations. They **DO NOT** go through the **thalamus**



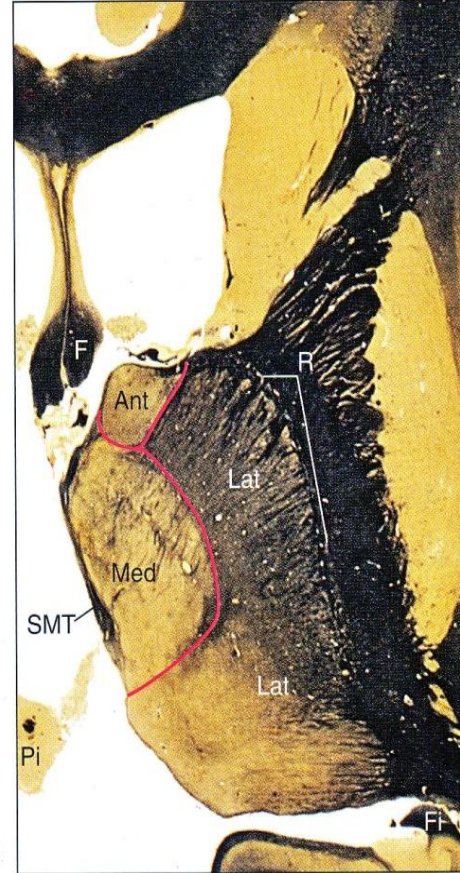
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A



B

Thalamic Nuclei

Thalamocortical radiations

Central sulcus

Thalamic nuclei

- CM Centromedian
- LD Lateral dorsal
- LP Lateral posterior
- MD Medial dorsal
- VA Ventral anterior
- VI Ventral intermedial
- VL Ventral lateral
- VPL Ventral posterolateral
- VPM Ventral posteromedial

From globus pallidus
and substantia nigra

Reticular nucleus (pulled away)

From cerebellum

Somesthetic from body
(spinothalamic tract and medial lemniscus)

Somesthetic from head (trigeminal nerve)

Internal medullary lamina

Intralaminar nuclei

Other medial nuclei

Midline (median) nuclei

Interthalamic adhesion

Pulvinar

Medial geniculate body (MGB)

Acoustic pathway

Lateral geniculate body (LGB)

Optic tract

F. Netter M.D.

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Thalamic Nuclei

Anterior (**A**)

Medial = Dorsomedial (**DM**)

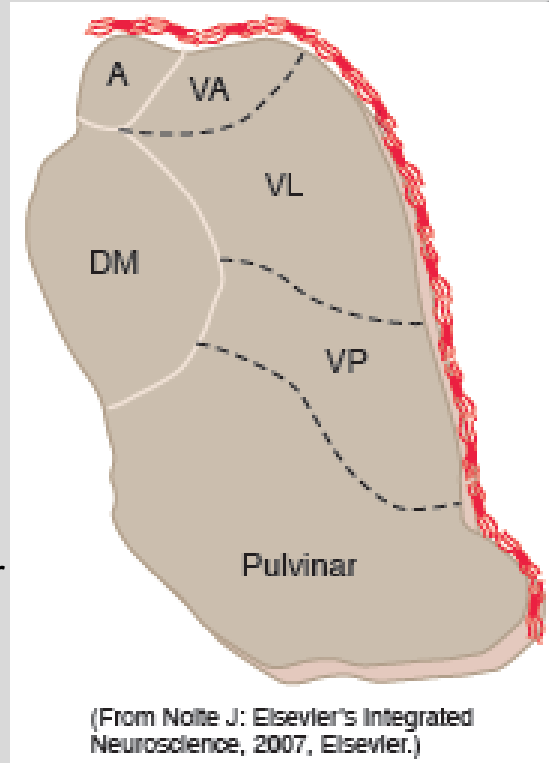
The **lateral division** is divided into several nuclei:

Ventral anterior (**VA**),

Ventral lateral (**VL**) and

Ventral posterior (**VPL/VPM**) nuclei.

Above VPL/VPM is the *lateral posterior nucleus* (**LP**), which merges with the **pulvinar**.



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Thalamic Nuclei

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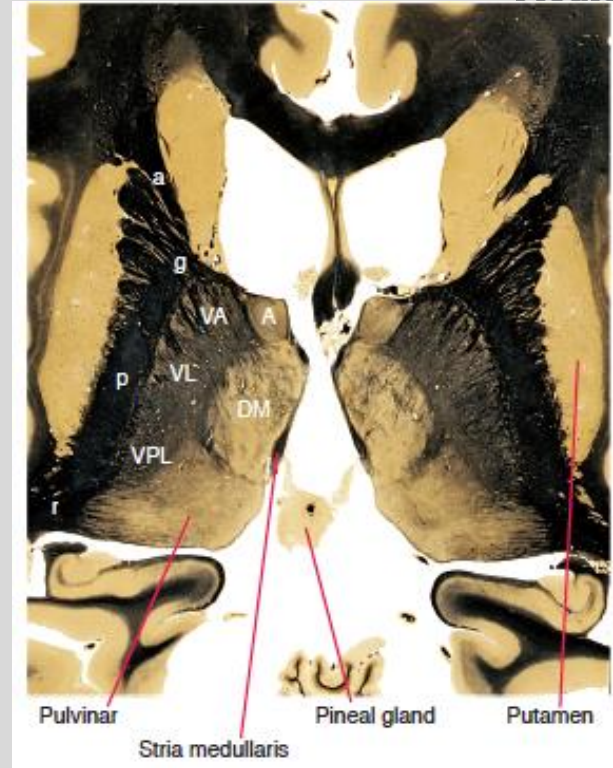
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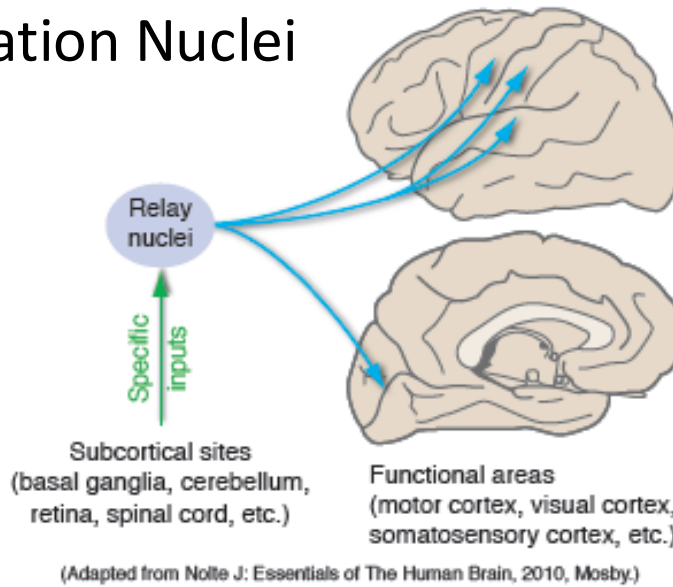


Thalamic Relay and Association Nuclei

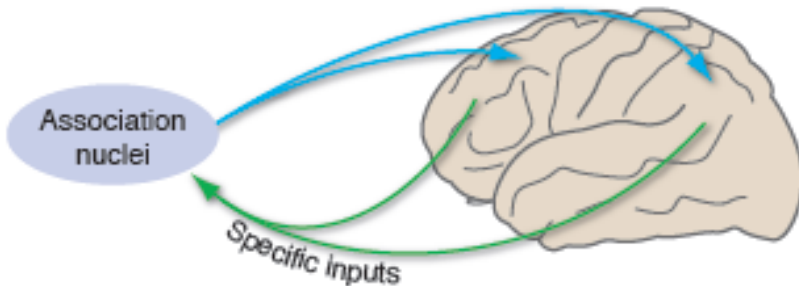
Two types of Nuclei in the Thalamus

Relay nuclei

Receive **specific inputs** via well-defined, fairly discrete input bundles, and **project to restricted cortical areas** with **single functions**



Association areas
(prefrontal,
parietal-occipital-temporal)



(Adapted from Nolte J: Essentials of The Human Brain, 2010, Mosby.)

Association nuclei

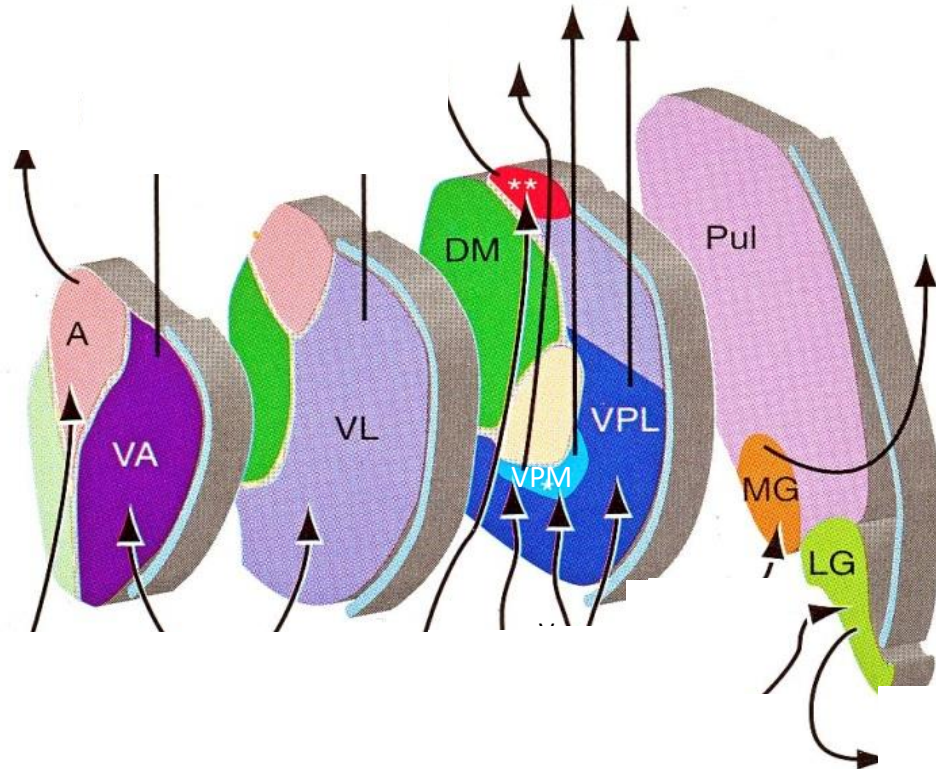
Receive inputs mainly from **association cortex**, and project **back** to the related areas of association cortex.

Thalamus Relay Nuclei

Divisions of the Thalamus;
Coronal view

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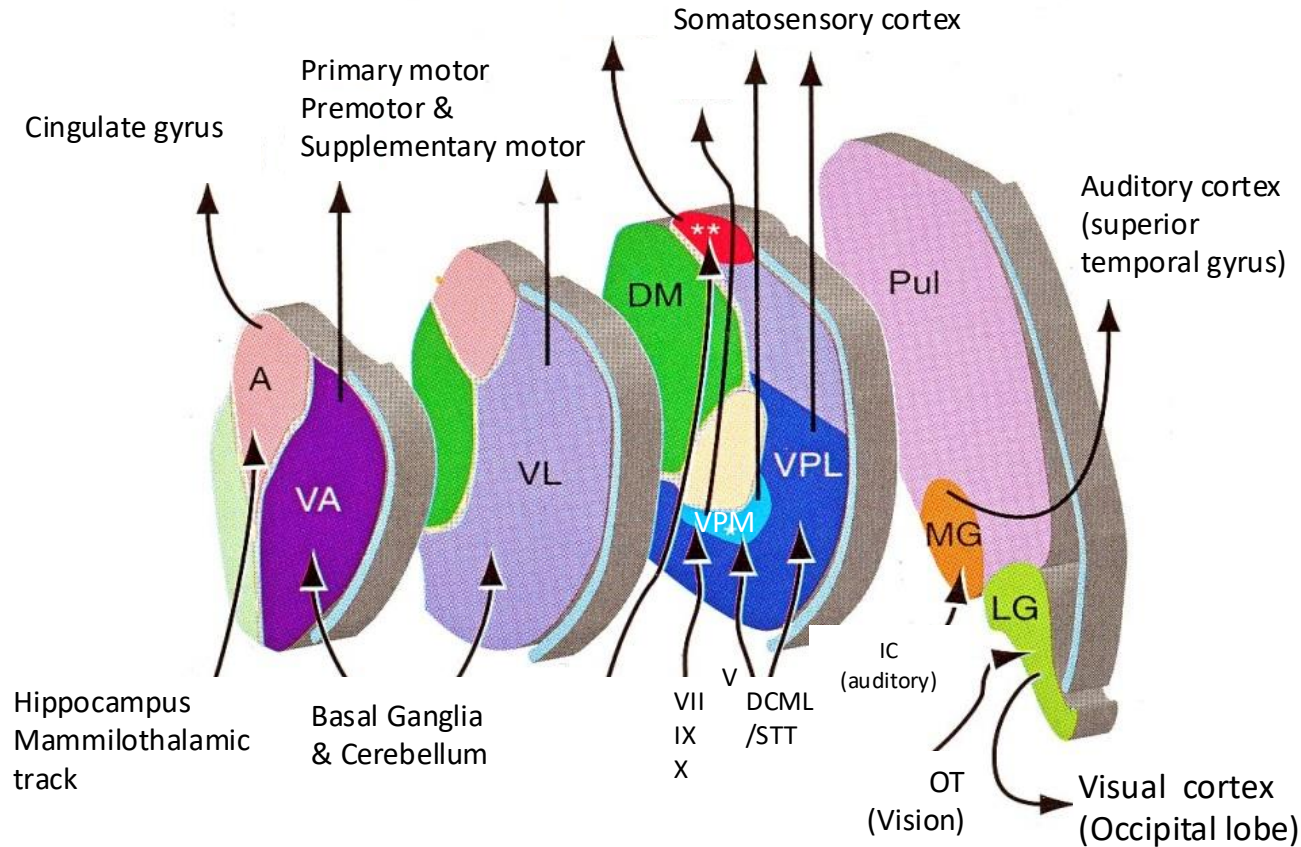


Thalamus Relay Nuclei

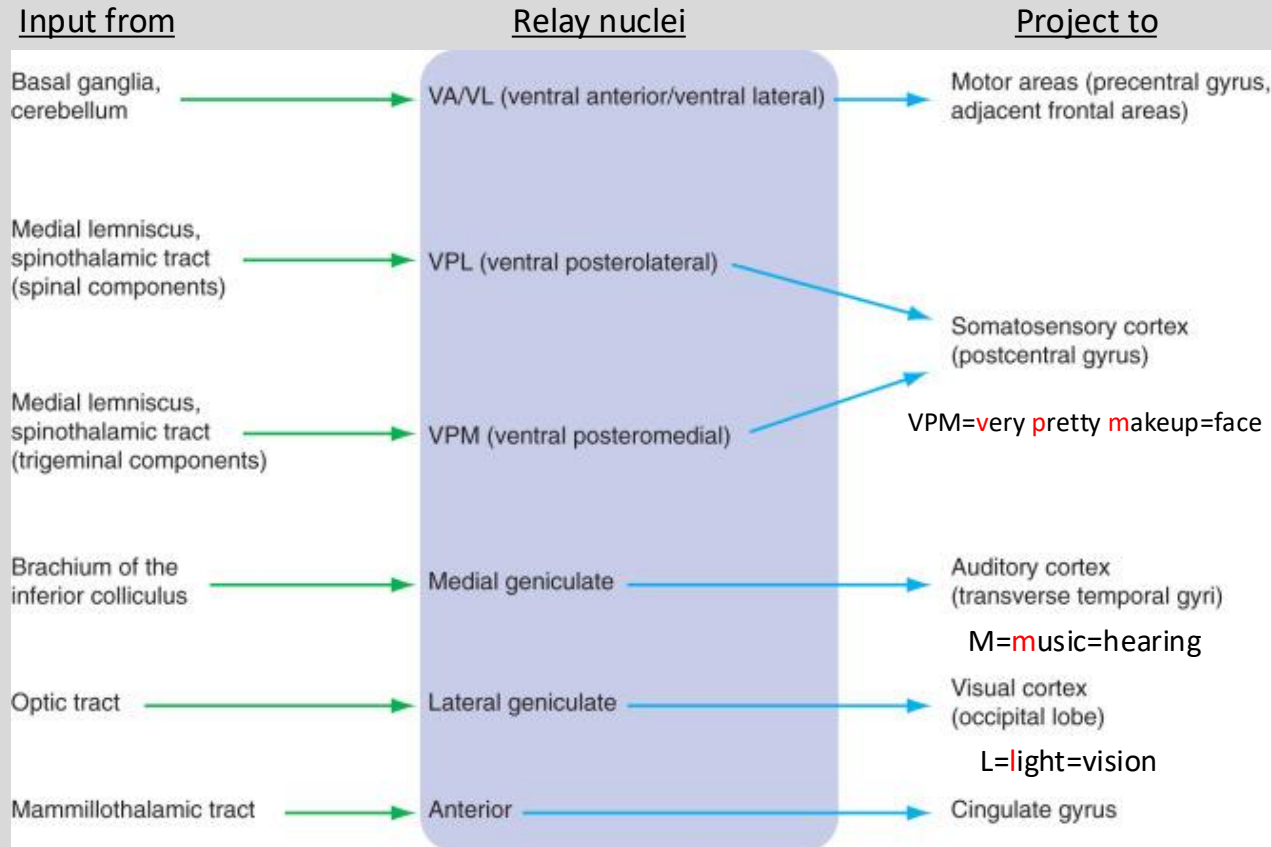
Divisions of the Thalamus;
Coronal view

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Thalamus Relay Nuclei



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Thalamocortical Pathways

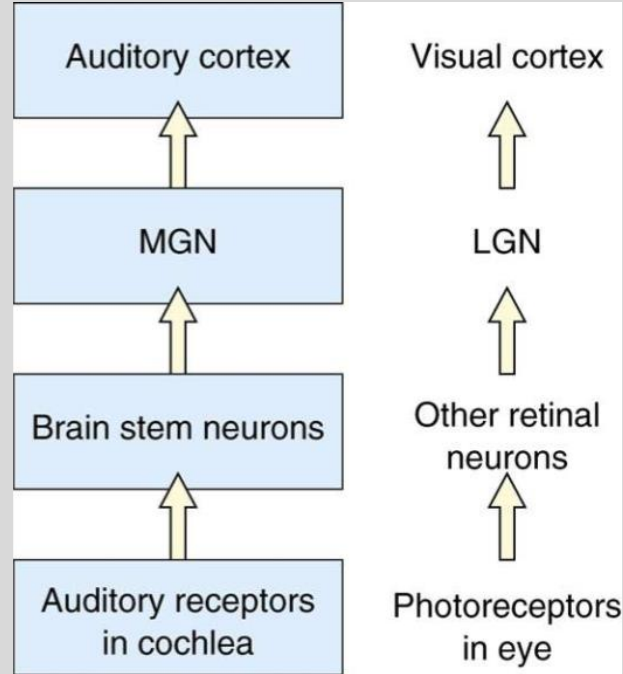
Largely similar circuits:

4) Neocortex – specific region & specific layers (**conscious perception**)

3) Thalamus – specific nucleus (**3rd order neurons**)

2) Intermediate centers – e.g. Brainstem (**2nd order neurons**)

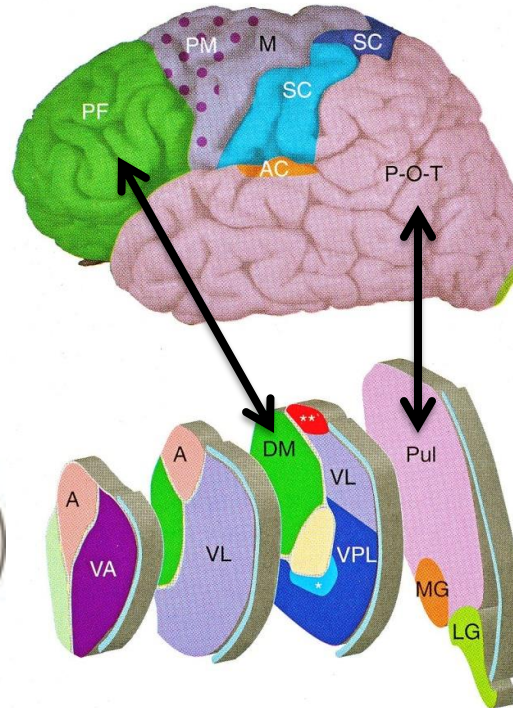
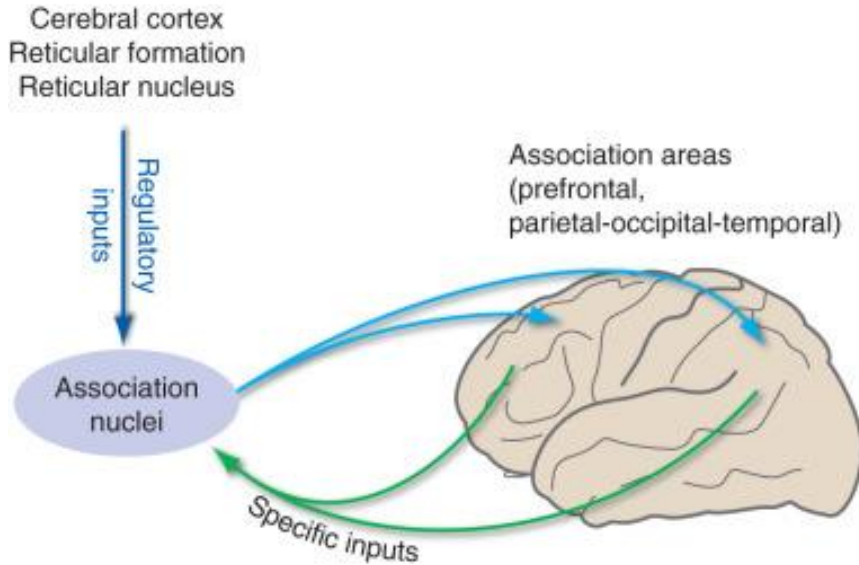
1) Start – activation of peripheral sensory receptors (**1st order neurons**)



Thalamic Association Nuclei

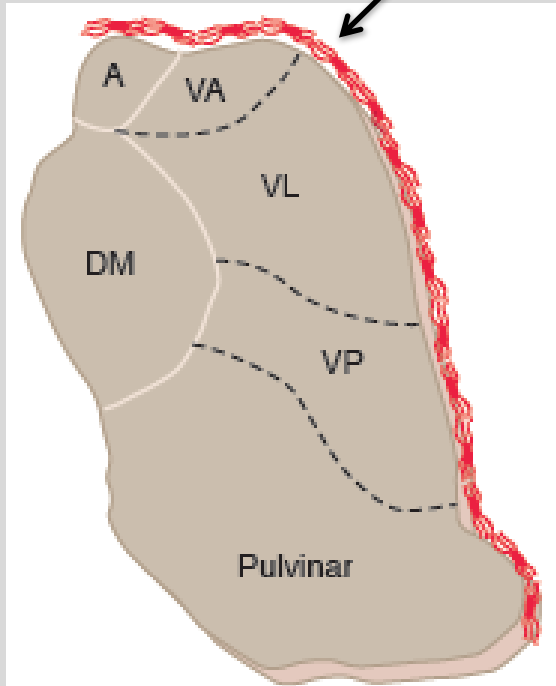
Receives information from the **association cortex**, and distribute it back to association cortex

- **Dorsomedial nuclei**
- **Pulvinar**



Thalamus Reticular Nucleus

Reticular nucleus



Samples the activity in fibers

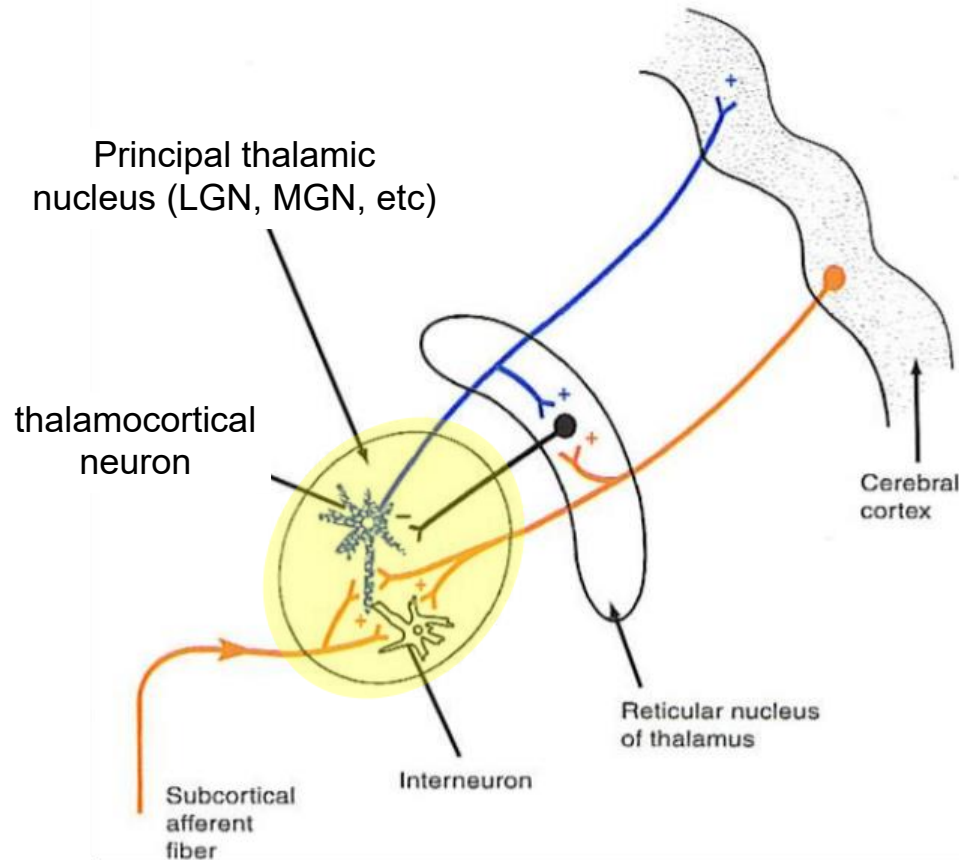
Sends **inhibitory (GABA)** projections **back to adjacent parts of the thalamus.**

- **Directs attention**
- **sleep-wakefulness cycles**

Thalamocortical Connections vs. Corticothalamic Connections

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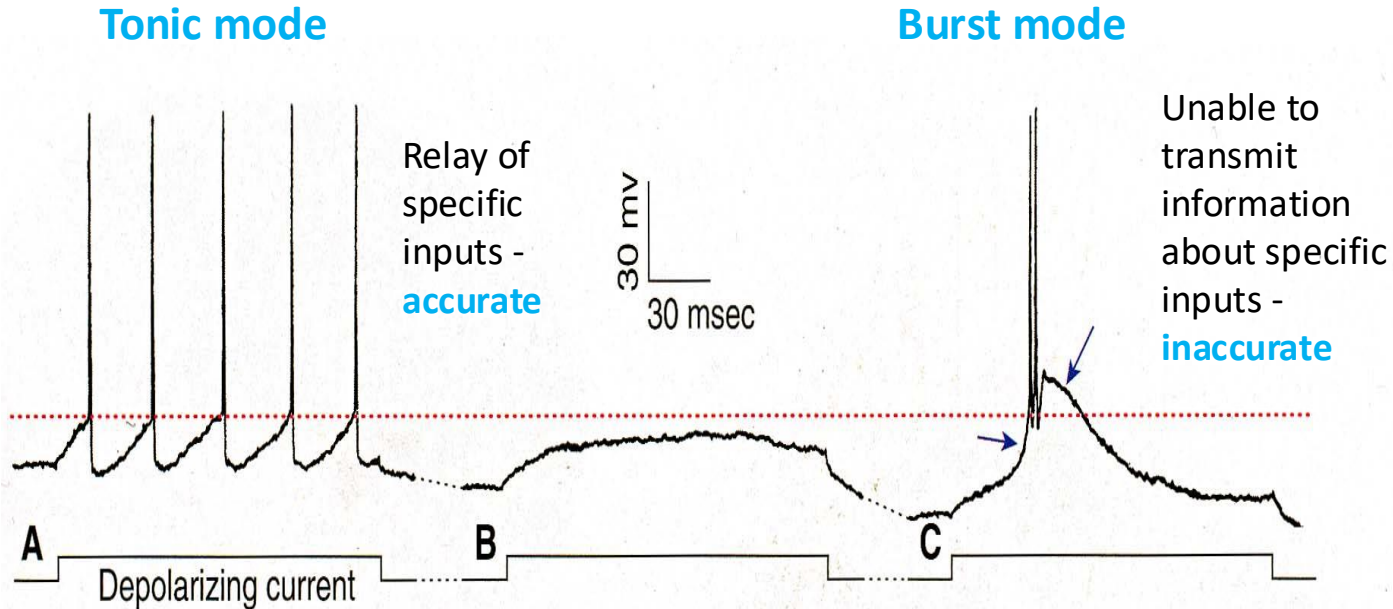


Two states

Thalamic projection neurons have *Two Physiological States*

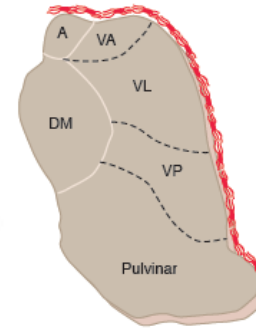
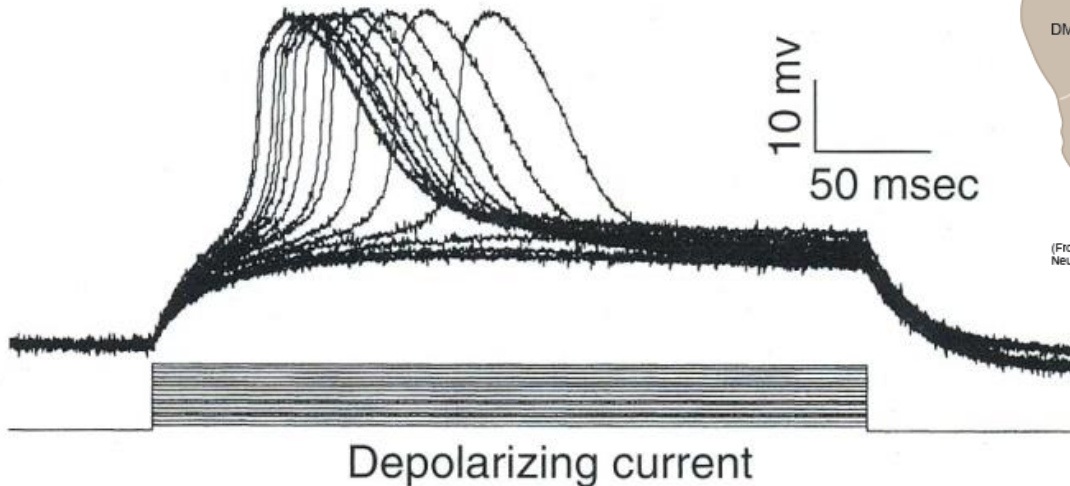
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Burst mode

- Long refractory period of calcium channels
- Reduction in action potentials
- Dynamic relay of information to the cortex during **sleep-wake**
- May underlie **absence epilepsy**



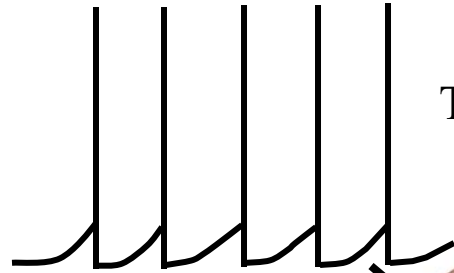
(From Nolte J: Elsevier's Integrated Neuroscience, 2007, Elsevier.)

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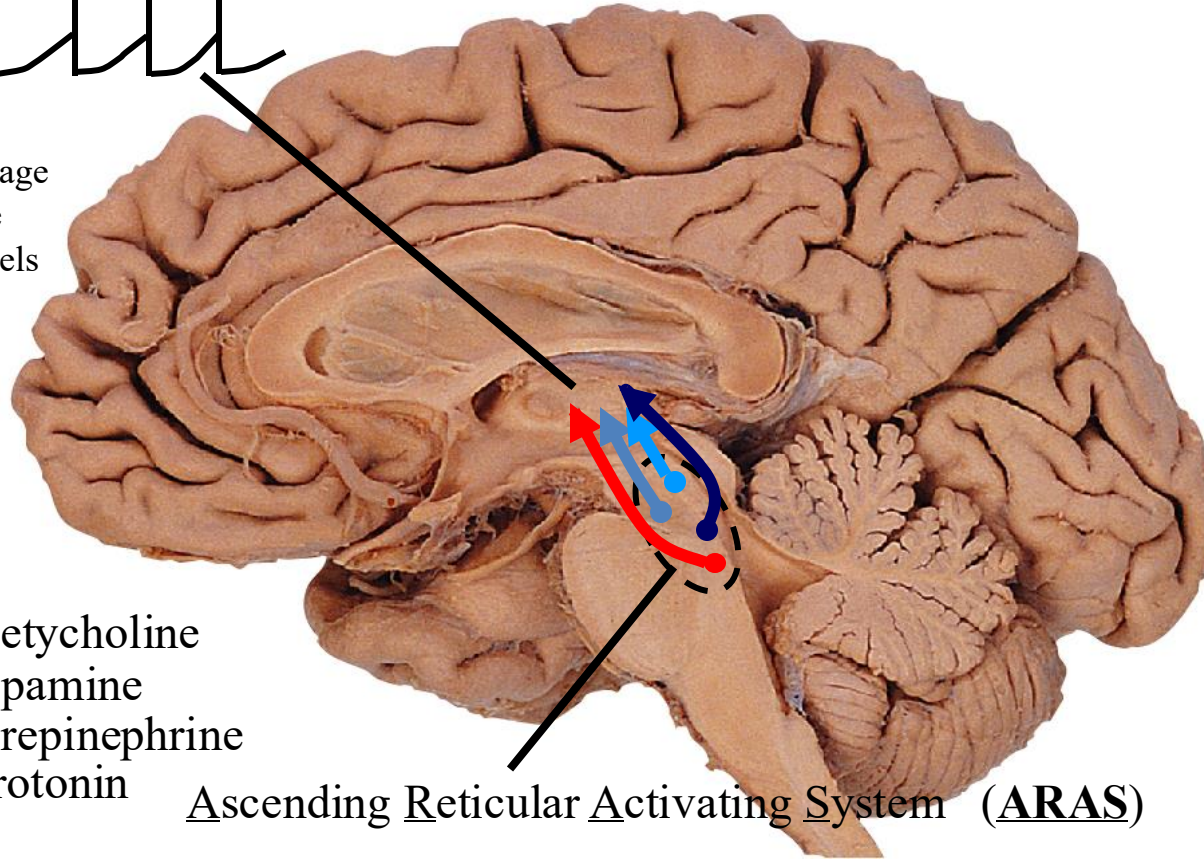
Awake

Tonic Mode



-65 mV

Special Voltage
Sensitive
 Ca^{+2} Channels
Inactive



- Acetylcholine
- Dopamine
- Norepinephrine
- Serotonin

Ascending Reticular Activating System (**ARAS**)

SWS Sleep

Burst Mode

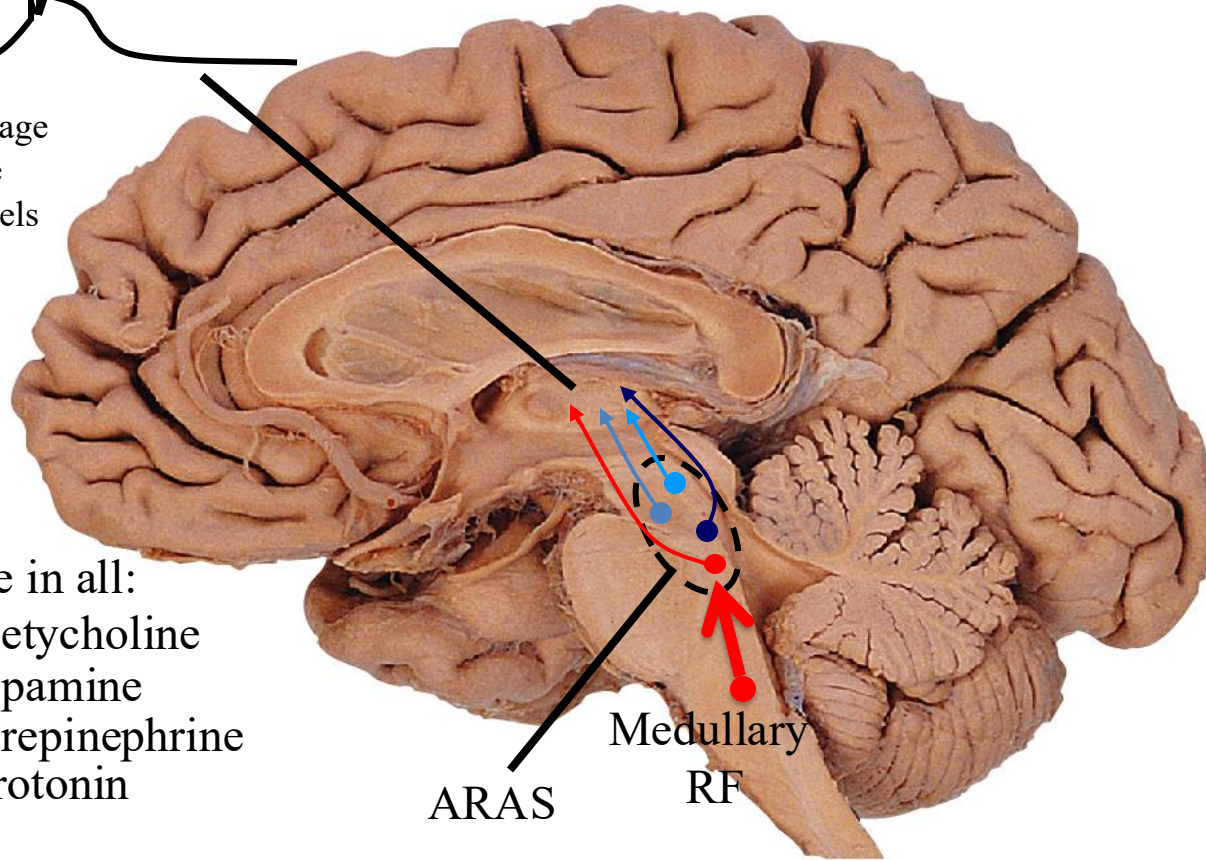
-80 mv

Special Voltage
Sensitive
 Ca^{+2} Channels
Active

SWS

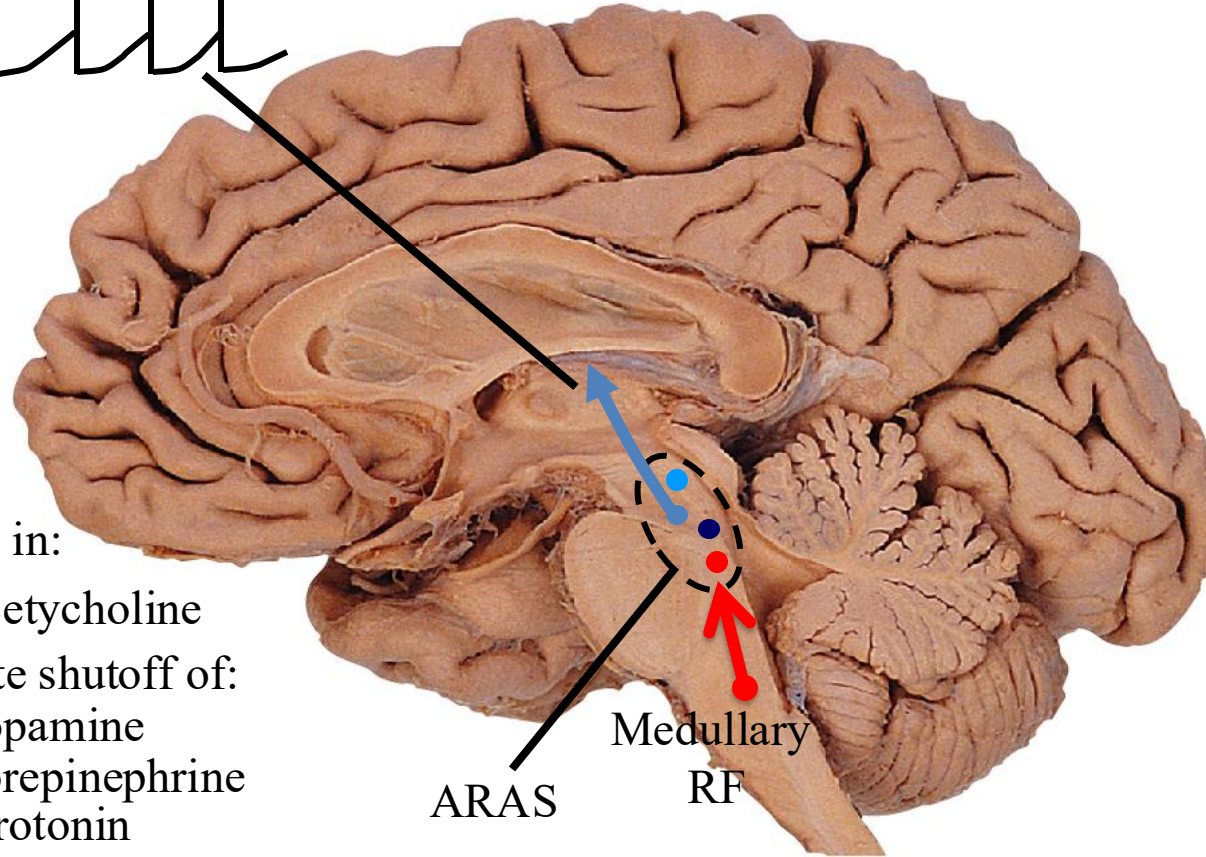
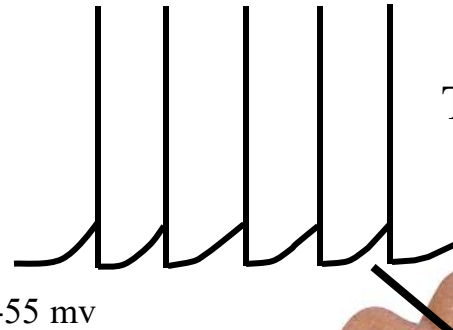
Decrease in all:

- Acetylcholine
- Dopamine
- Norepinephrine
- Serotonin



REM Sleep

Tonic Mode



REM

Increase in:

● Acetylcholine

Complete shutoff of:

● Dopamine

● Norepinephrine

● Serotonin

ARAS

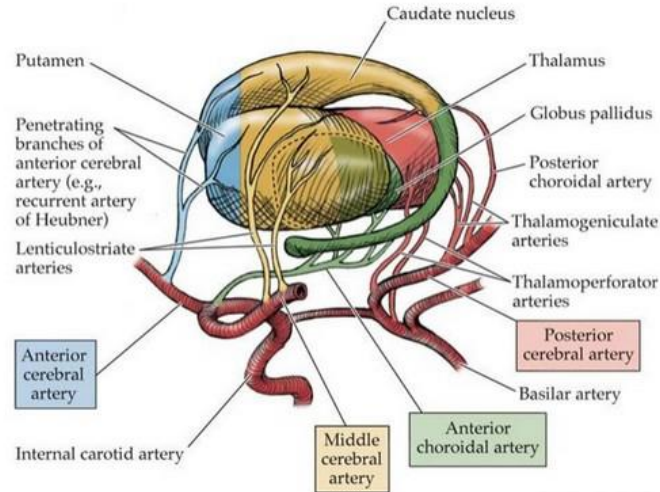
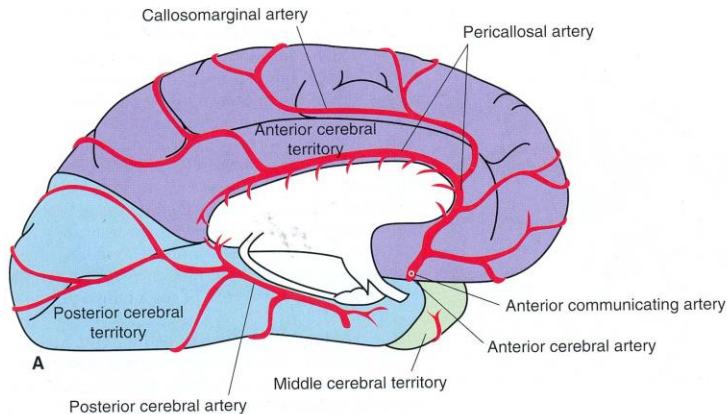
Medullary
RF

Blood Supply to the Thalamus

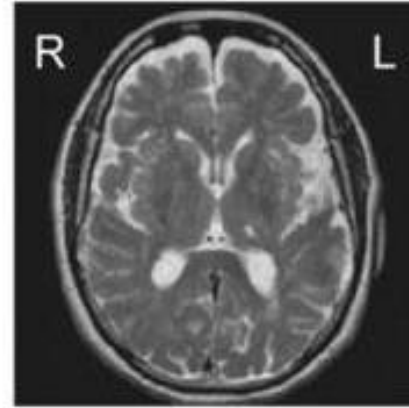
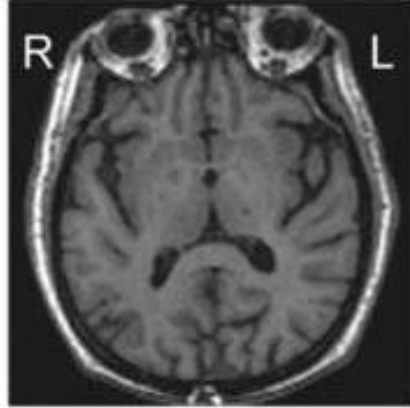
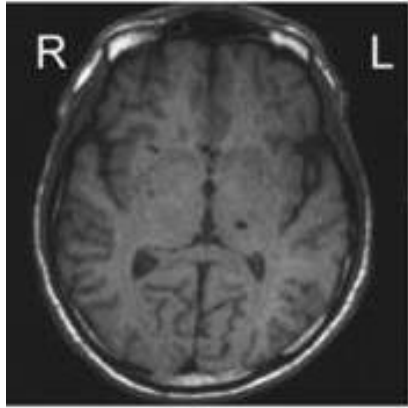
Posterior Cerebral Artery: small branches supply most of the thalamus

Posterior communicating artery

Anterior choroidal artery: lateral geniculate body

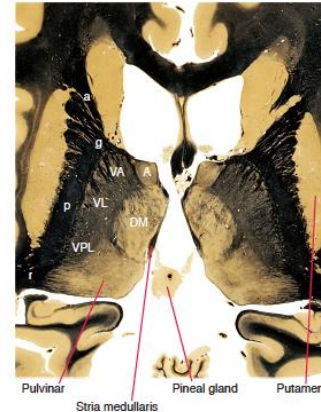


Ventral/Posterior Thalamic Lesions and Central Post Stroke Pain Syndrome (Dejerine–Roussy syndrome)



Syndrome of neuropathic pain due to thalamic lesions.
Occurs in ~10% of stroke patients

- Initially **paresthesias** (numbness & tingling)
- Followed in weeks to months by **allodynia** (ordinarily painless stimuli cause pain)
- Pain described as **temperature** (hot, burning, cold)
- Pain described as **mechanical** sensation (dull, aching, squeezing)

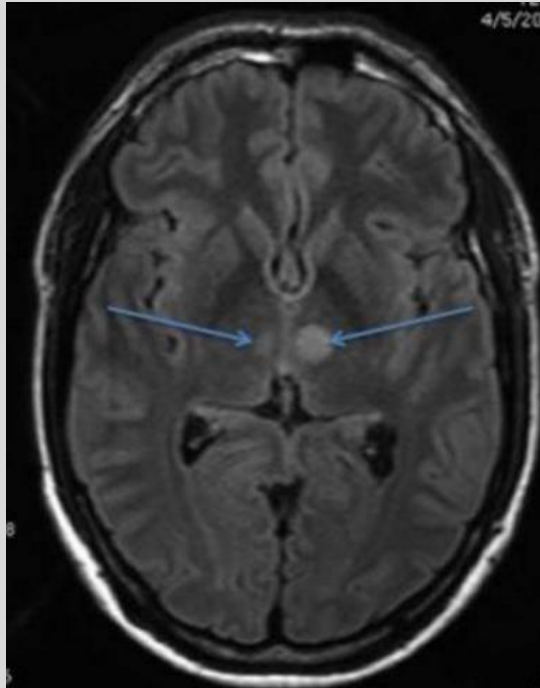


Case

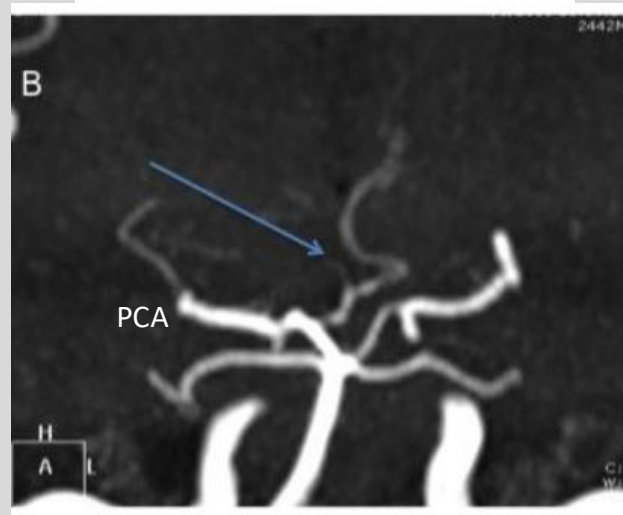
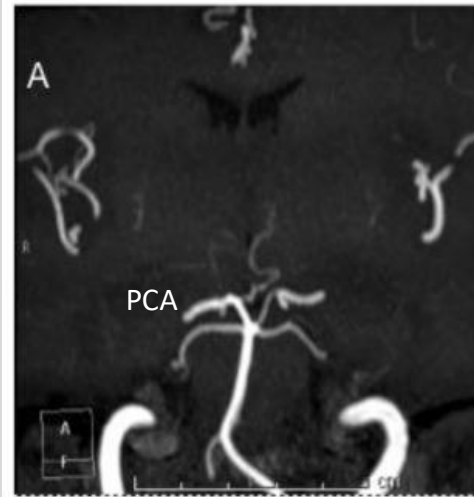
31-year-old man presented to ED 6 days after experiencing visual, speech & gait disturbances. The symptoms had improved over those 6 days, his sole remaining complaint is impaired vision. The only abnormality of physical and neurological examinations is dysarthria. All lab blood test are negative, including a profile for hypercoagulable state. He has a history of viral meningitis & seizures as a child. Audiovisual EEG showed no abnormalities.

Pause the video here...

- Non-contrast MRI and MRA



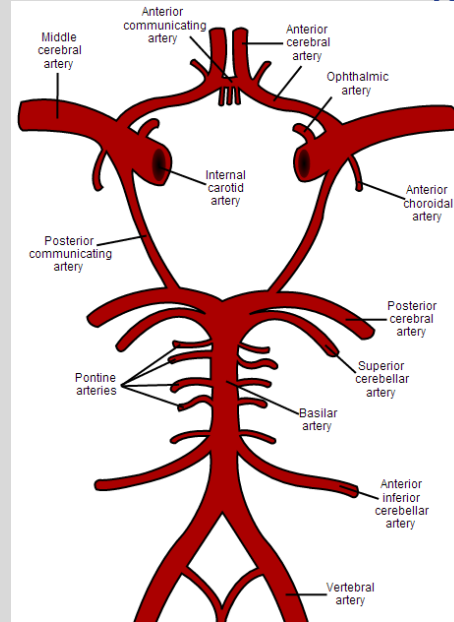
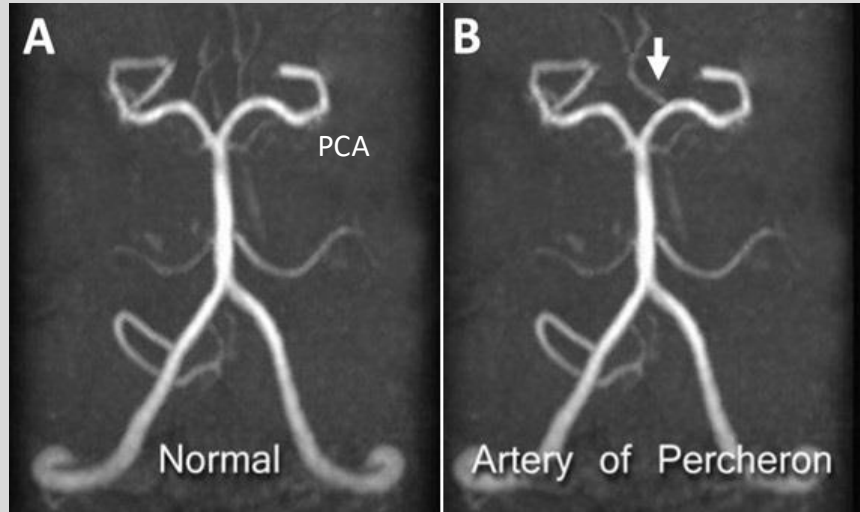
J Radiol Case Rep. 2013 Jul; 7(7): 7–14.



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- Diagnosis: Occlusion of **Artery of Percheron**



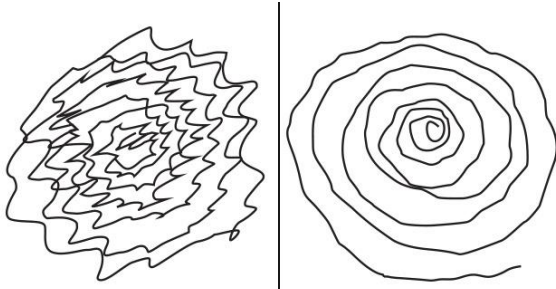
- Occlusion of a rare vascular variant → **bilateral paramedian thalamic infarcts** with or without midbrain involvement
- Most common cause → cardioembolism
- Favorable prognosis

Clinical Research

The NEW ENGLAND JOURNAL of MEDICINE

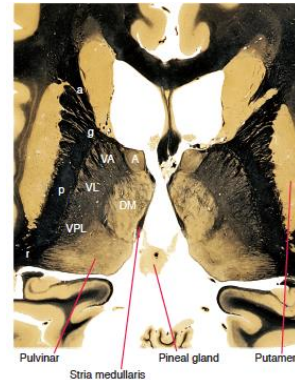
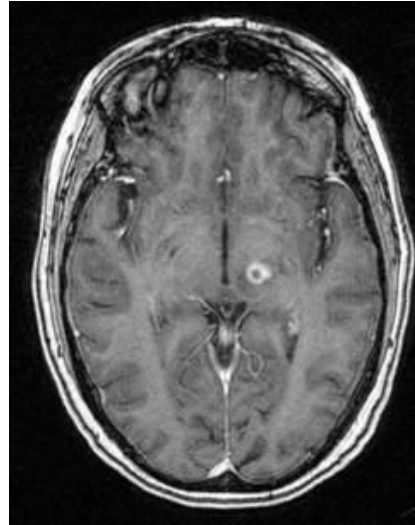
ORIGINAL ARTICLE

A Randomized Trial of Focused Ultrasound Thalamotomy for Essential Tremor



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- MRI-guided focused ultrasound thalamotomy reduced hand tremor in patients with essential tremor.
- Side effects included sensory and gait disturbances.



N Engl J Med 2016;375:730-9.

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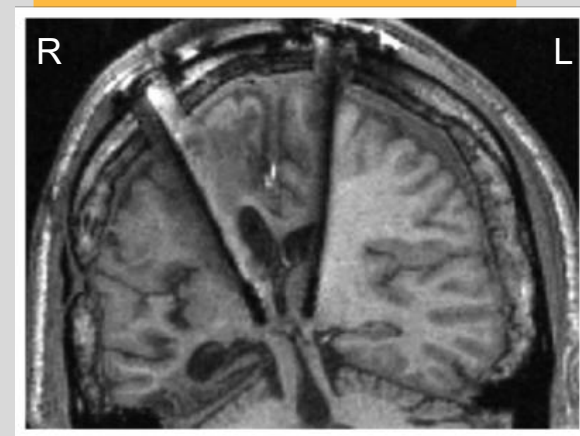
Stimulation of central thalamus revives minimally conscious patient

Background

- 38 yo right-handed man sustained a closed head injury following an assault
- Suffered blunt trauma to right frontal region → bilateral subdural hematomas (right > left).
- Initial Glasgow Coma Scale: 3 (deep coma)
- Right temporal craniotomy was performed → evacuation of hematoma
- Post-op patient remained behaviorally unresponsive for at least 2 months

Findings

- Deep brain stimulation (DBS) of thalamus produced acute changes in arousal including
 - Increased heart rate, well sustained eye opening & rapid bilateral head turning to a voice
- On day ~143 patient showed first instances of functional object use and intelligible verbalization (no previous episodes were observed on previous 33 evaluations/6 mo.)
- Behavioral improvements temporally coincided with onset of DBS and did not emerge until 160 days after initiation of rehabilitation program, indicating they were partially attributable to DBS.

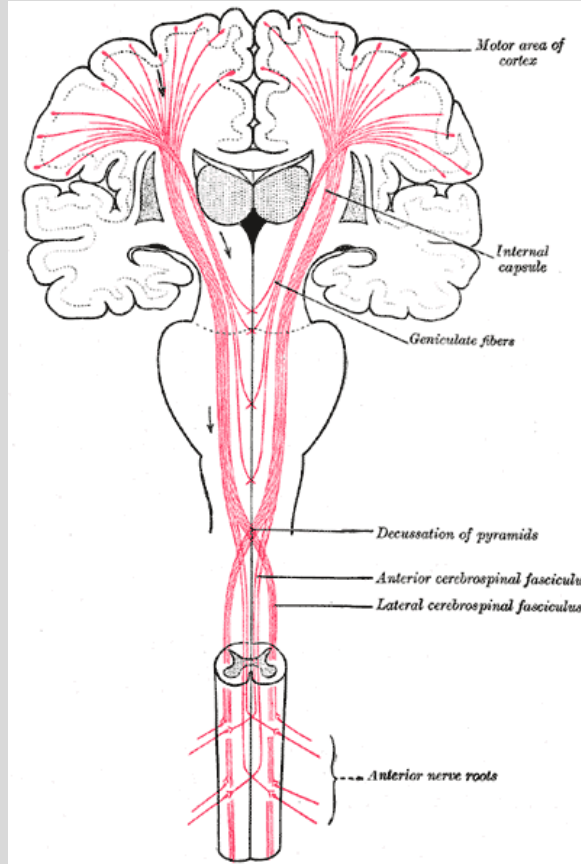


MRI reveals the position of stimulating electrodes in the thalamus

Schiff et al. 2007

Internal Capsule

- Thalamocortical neuron pathways and other ascending tracts to the cortex
- Pathways for major descending long tracts from cortex to destinations



Internal Capsule Pathways

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5 pathways

1. Anterior Limb

Between lenticular and caudate nucleus

2. Genu

3. Posterior Limb

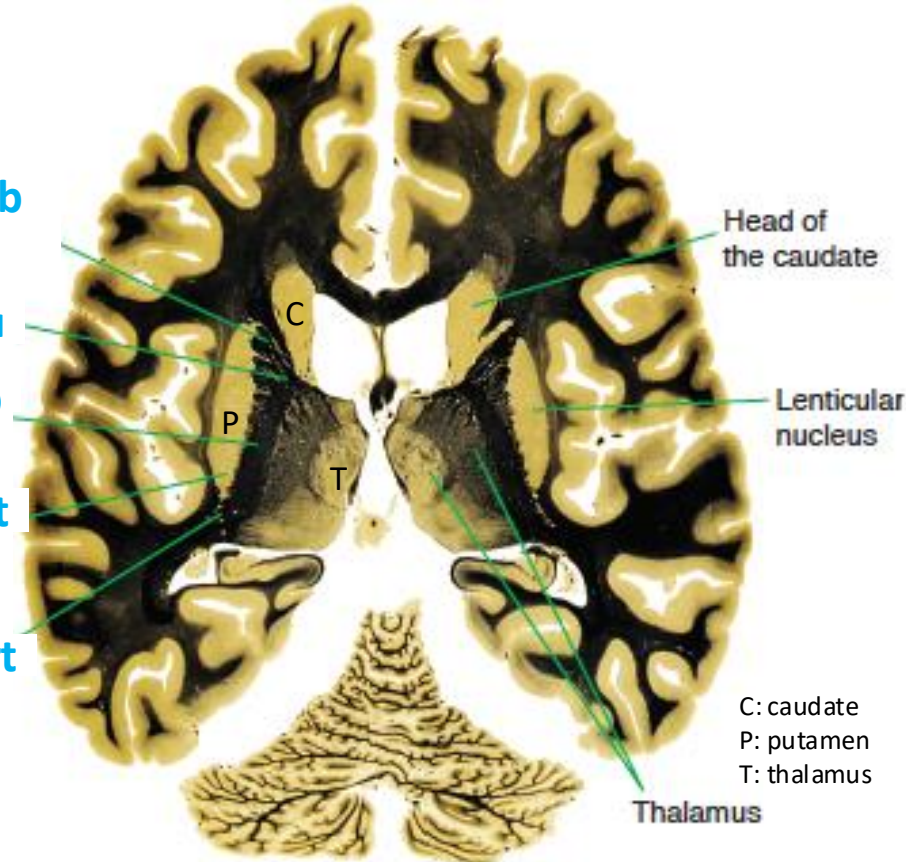
Between lenticular and thalamus

4. Sublenticular part

Under the lenticular nucleus

5. Retrolenticular part

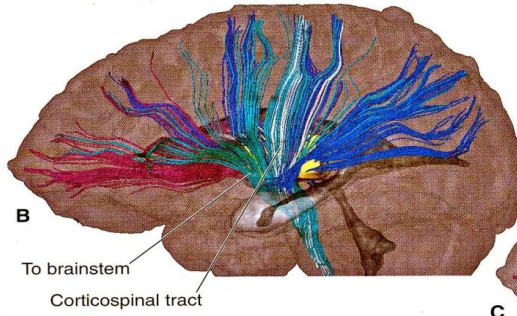
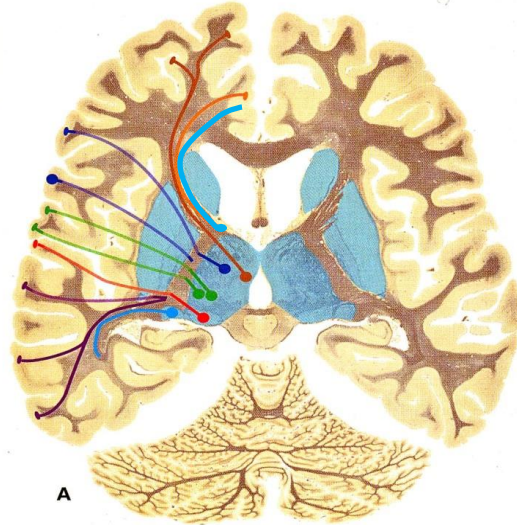
Behind the lenticular nucleus



Internal Capsule

The neuronal axons in **each of these areas** carry markedly different information

All color coded 😊



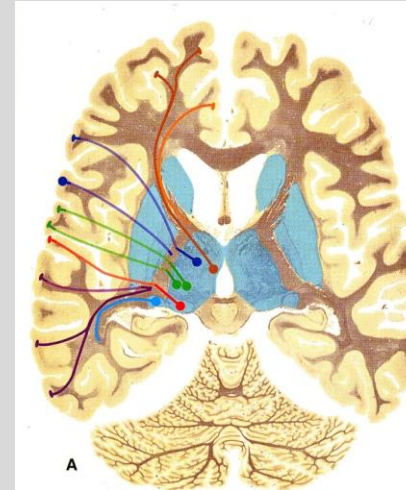
- Anterior nucleus → cingulate gyrus
Anterior limb IC
- DM → prefrontal cortex
Anterior limb IC
- VA/VL → motor areas
Motor areas → brainstem, spinal cord
Posterior limb IC
- VPL/VPM → somatosensory cortex
Posterior limb IC
- MGN → auditory cortex
Sublenticular part IC
- Pulvinar/LP → parietal-occipital-temporal cortex
Retrolenticular & *Sublenticular IC
- LGN → visual cortex
Retrolenticular & *Sublenticular IC
- Corticobulbar tracts → brainstem
Genu IC

Content of Internal Capsule

Portion	Origin	Output Destination	Other names
Anterior Limb	Anterior nucleus	Cingulate gyrus	
	DM	Prefrontal cortex	
Genu	Motor cortex	Brainstem	Corticobulbar tract
Posterior Limb	Motor cortex	Spinal cord	Corticospinal tract
	Motor cortex	Brainstem	corticopontine tract
	VPL/VPM	Postcentral gyrus	Somatosensory radiation
Retrolenticular part	Pulvinar	Association cortex	
	LGN	Visual cortex	Optic radiation
Sublenticular part	LGN	Visual cortex	Optic radiation
	MGN	Auditory cortex	Auditory radiation

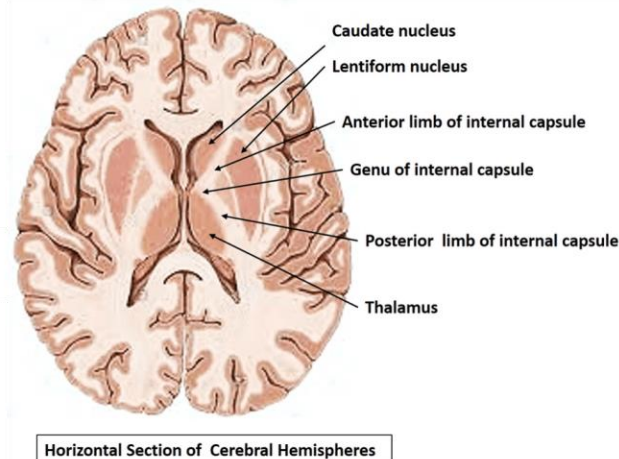
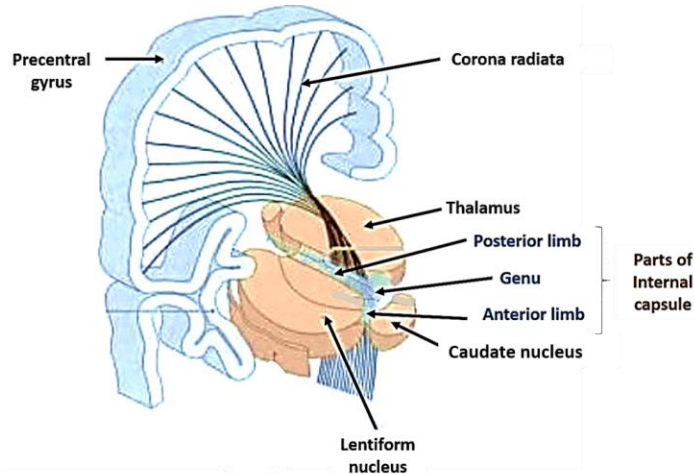
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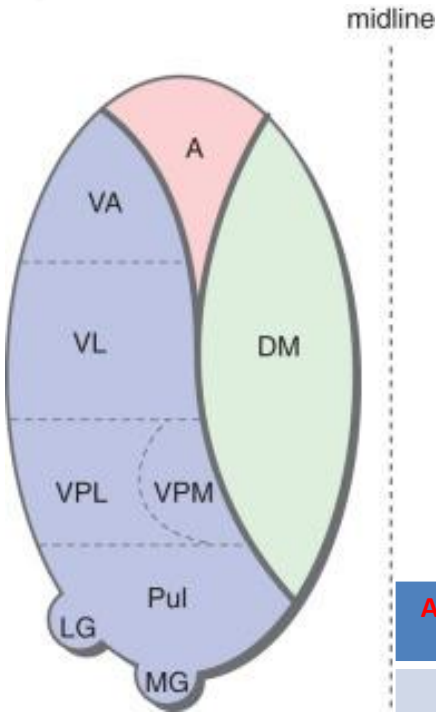
Blood Supply of Internal Capsule

IC Limb	Major Artery	Perforating Branches
Anterior IC	MCA ACA	Lenticulostriate a. Recurrent a. of Heubner
Posterior IC	MCA ICA	Lenticulostriate a. Anterior choroidal a.



Test yourself

Thalamus Summary



Relay Nucleus	Input (From)	Info	(To) Output Destination
VPL			
VPM			
LGN			
MGN			
VA/VL			
A			

Association Nucleus	Input (From)	Info	(To) Output Destination
DM			
Pulvinar			

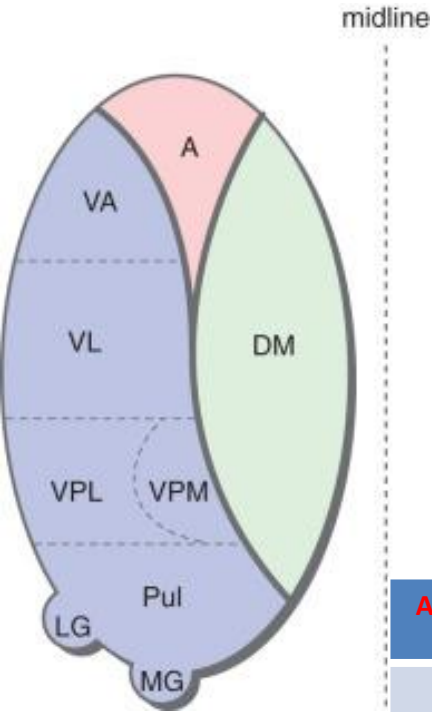
YORK INSTITUTE
TECHNOLOGY

Department of Osteopathic
Physiotherapy

Thalamus Summary

NEW YORK INSTITUTE
TECHNOLOGY

College of Osteopathic
Physicians and Surgeons

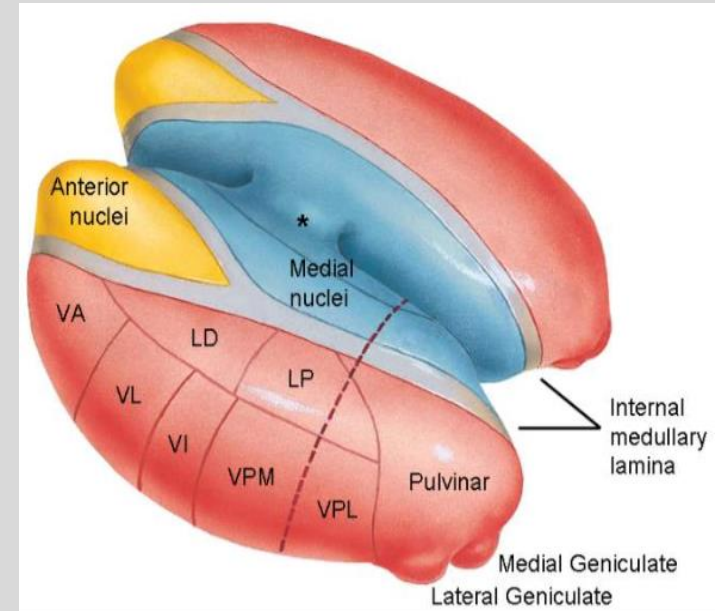


Relay Nucleus	Input (From)	Info	(To) Output Destination
VPL	Spinothalamic & dorsal columns/ medial lemniscus	Pain, temperature; pressure, touch, vibration, proprioception	Somatosensory cortex
VPM	Trigeminal & gustatory pathway	Face sensation, taste	Somatosensory cortex, insula/ frontal operculum
LGN	Optical tract	Vision	visual cortex (occipital lobe)
MGN	Superior olive and inferior colliculi of tectum	Hearing	Auditory cortex (temporal lobe)
VA/VL	Basal ganglia, cerebellum	Motor	Motor Cortex
A	Mammillary body hippocampus	Memory	Cingulate Cortex

Association Nucleus	Input (From)	Info	(To) Output Destination
DM	Prefrontal assoc. cortex	General arousal, modulation of thalamic input to the cortex	Prefrontal assoc. cortex
Pulvinar	Parietal, occipital, temporal assoc. cortex	General arousal, modulation of thalamic input to the cortex	Parietal, occipital, temporal assoc. cortex

Thalamic functional organization

- **Lateral** nuclear group – located ventrally; general and special senses
- **Anterior** nuclear group – part of limbic system
- **Medial** nuclear group – mood & emotion
- **Intralaminar** nuclei – together with medial nucleus for cortico-thalamo-cortical connections to regulate cognition and arousal
- **Reticular nucleus** – the only inhibitory nucleus in thalamus to regulate cerebral cortex activation (together with ARAS)



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